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A Model for Understanding the Effects of Sediment Dynamics on Oyster Reef Development

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Abstract Previous field studies have demonstrated that sedimentation is an important factor that can limit oyster reef growth and restoration success. High relief reefs are more productive and resilient than low relief reefs, in part, because increasing reef height reduces sedimentation and enhances oyster growth. In this study, we investigated the relationship between initial reef height and reef development using a simple model. The model contains three coupled differential equations that describe changes in oyster volume, shell volume, and sediment volume per unit area of reef with time. The model was used to investigate how parameters such as flow speed, sediment grain size, and food concentration affect reef survival and final reef height. Whether or not a reef survives depends primarily on the shape of the sediment concentration profile relative to the initial reef height. Over a long time period, three different steady-state reef heights are possible, depending on the environmental parameters and initial reef height: (1) If growth outpaces sedimentation, the reef achieves the maximum possible height, which is independent of sedimentation parameters; (2) if deposition outpaces growth and the shear stress does not exceed the critical shear stress, the reef is buried in sediment and dies; and (3) if deposition outpaces growth and the shear stress exceeds the critical shear stress, a reduced steady-state height is achieved that depends on both growth and sedimentation parameters. The model can be used to assess the ways in which measurable environmental parameters affect reef restoration success.

Keywords Oyster reef · Sediment · Restoration · Model · Estuary

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Introduction

Oysters are ecosystem engineers that affect physical, chemical, and biological processes in estuaries and perform important ecological functions. In addition to their importance as a fishery, the hard biogenic substrate provides important habitat for numerous species of fish and invertebrates (Peterson et al. 2003). Oysters also help maintain water quality by filtering particulate matter including phytoplankton, organic matter, and sediment from the water column (Kellogg et al. 2013). In this process, nutrients are removed from the water column and deposited on the seafloor where they are accessible to submerged aquatic vegetation and benthic microalgae. Additionally, the presence of oyster reefs reduces erosion and preserves adjacent wetland environments (Meyer et al. 2008). Overall, the health and productivity of the estuarine environment is improved with the presence of the oysters.

In estuaries along the Atlantic coast, fishery landings of the oyster *Crassostrea virginica* (Gmelin 1791) have declined more than 90 % relative to the historic peaks of the 1900s as a result of overfishing and anthropogenic pollution (Grabowski and Peterson 2007). Oyster population loss has contributed to the decline of ecosystem health in these systems. In the Chesapeake Bay, degradation of water quality has been demonstrated to be both a cause and an effect of oyster population decline. Oysters in the Chesapeake were estimated to filter the volume of

the entire water body in 3.3 days in the later 1800s. By 1988, oyster populations had decreased so much that it would take 325 days for the oysters to filter the entire volume of the Chesapeake, and overall ecosystem health had simultaneously declined (Newell 1988).

Over the past few decades, there has been a push to restore oyster reefs, both to improve oyster harvests and recover ecosystem services. Oyster reef restoration typically involves building three-dimensional substrate (e.g., shell, limestone, concrete) structures that imitate natural oyster reef formations to promote the recruitment of oyster larvae. Restored reefs have been constructed in both the intertidal and subtidal zones of estuaries where the maximum water depth is <3–5 m because these are the areas where oyster reefs occur naturally. The designs, layouts, and locations of restoration projects vary regionally due to differing environmental conditions, resource availability, and project objectives. Despite the broad scale of restoration efforts, reef construction projects have failed to yield consistent positive results (Powers et al. 2009). Failures in reef restoration efforts can usually be attributed to smothering by sediment, hypoxia and anoxia, poor larval recruitment, and disease (Powers et al. 2009).

Although many reef restoration efforts have been unsuccessful, some of these projects have achieved good results. For example, a restoration effort in the Great Wicomico River in Virginia created the largest metapopulation of oysters restored in any nation worldwide (Schulte et al. 2009). Both low and high relief reefs were constructed during this restoration effort. The high relief reefs experienced less sedimentation and were vastly more successful than the low relief reefs. They had an oyster density that was five times larger after 3 years and were more resilient to disturbance. This dichotomy between the success of low and high relief reefs suggests the outcome of reef restoration efforts may be strongly influenced by sediment dynamics.

High rates of sedimentation negatively affect oyster growth. When sediment concentrations are high, the gills of the oysters become clogged and the oysters suffocate and die (Adams et al. 1995). Additionally, the oysters must expend energy to filter sediment from the water column, which increases the rates of disease and mortality (Barillé et al. 1997). Consequently, high rates of sedimentation negatively impact reef restoration efforts (Powers et al. 2009).

Suspended sediment is typically more concentrated in the lower water column. The balance between downward sediment settling and upward turbulent diffusion leads to a concentration profile that decays exponentially with height above the bottom (van Rijn 1993). As a result, the negative effects of sedimentation on oyster reefs decrease as reef height increases because less deposition occurs higher in the water column (Schulte et al. 2009; Stokes et al. 2012). Additionally, as the reef becomes covered by sediment, fewer oysters are able to filter, leading to an increased concentration of sediment in the water above the reef, which increases the rate of sedimentation on the reef.

The goals of this study were (1) to extend an existing mathematical model for the effects of sedimentation on oyster reef development by improving the representations of physical processes and incorporating real environmental parameters, (2) to use the model to understand how physical and biological processes interact to affect reef survival and productivity, and (3) to investigate how different attributes of a site affect reef restoration success. Our focus was the oyster *C. virginica*.

The starting point for the study was a simple mathematical model developed by Jordan-Cooley et al. (2011); henceforth, JC11. The model consists of three coupled ordinary differential equations that represent oyster reef development and sediment accumulation through time. The set of equations have a bifurcation point, a threshold initial reef height above which an oyster reef will survive and below which it will not. In JC11, simple algebraic expressions were used to represent processes such as oyster growth, sediment deposition, and erosion so that the mathematical properties of the equation set could be explored. While the behavior of these terms is intuitive, their formulations do not correctly represent the physical processes or include real environmental parameters.

In this study, we introduce new forms for terms of the JC11 model that better represent these processes. Specifically, (1) we account for volume fractions occupied by oysters, shell, and sediment; (2) we introduce a new density dependent growth term that represents particulate food uptake by filter feeders in a turbulent boundary layer; and, (3) we modify the equation for sediment accumulation to better represent erosion and deposition processes. The new set of equations contains measurable parameters such as water velocity, sediment grain size, and ambient food concentration. The revised model is used to investigate how different processes interact to affect oyster reef survival and productivity and to identify site characteristics that are optimal for oyster reef restoration success.

Model Development

The Original Model

The model developed by Jordan-Cooley et al. (2011) contains three coupled ordinary differential equations (Eqs. 1, 2, and 3) that represent changes in the volumes of live oyster (O), shell (B), and sediment (S) per unit

plan area with respect to time (t) as the oyster reef develops,

$$\frac{dO}{dt} = \overbrace{rOf(1-O/K)}^{\text{Growth}} - \overbrace{\mu f O - \varepsilon(1-f)O}^{\text{Mortality}} \quad (1)$$

$$\frac{dB}{dt} = \overbrace{\mu f O + \varepsilon(1-f)O}^{\text{Mortality}} - \overbrace{\gamma B}^{\text{Shell Degradation}} \quad (2)$$

$$\frac{dS}{dt} = -\overbrace{\beta S}^{\text{Erosion}} + \overbrace{\chi G e^{-FO/\chi G}}^{\text{Deposition}} \quad (3)$$

The solution of this equation set gives the evolution of oyster, shell, and sediment volumes per unit plan area with time. The oyster reef simulated by this model is assumed to be subtidal, meaning that the top of the reef is always completely submerged.

Processes represented in the model are depicted in Fig. 1, and model parameters are summarized in Table 1. Oyster volume increases due to growth and decreases due to mortality from predation, disease, and smothering by sediment. As oysters die, they are converted to shell, increasing the shell volume. Shell volume decreases as the shell degrades due to weathering. Change in sediment volume occurs due to deposition and erosion. All volumes are represented as volume (m^3) per square meter of seafloor. Because the change in oyster volume is affected by the rate of oyster mortality due to sedimentation, and the rate of sediment deposition onto the reef is determined by the height of the reef, this system of equations represents the interactions between oyster reef growth and sedimentation (Fig. 1).

In JC11, oyster reef growth is modeled using a logistic growth expression that includes an instantaneous growth rate (r) and a carrying capacity (K). Oyster mortality is composed of two terms, natural mortality (μ) and an elevated mortality that occurs when oysters are buried by sediment (ε). Shell degradation rate is modeled as a constant fractional decay rate (γ). In the sedimentation term, erosion is modeled as the loss of a constant proportion of the sediment on the reef per unit time (β). Deposition rate decreases exponentially with increasing reef height and oyster volume and contains a maximum deposition rate (χ), a modification to sediment deposition that decays exponentially as the reef height increases (G), and a filtration rate (F) that represents the volume of sediment filtered per unit time per unit live oyster volume:

$$G = e^{-\eta(O+B)} \quad (4)$$

$$F = F_0 \frac{\chi G}{(\chi G)_0} \exp\left(1 - \frac{\chi G}{(\chi G)_0}\right) \quad (5)$$

Here, η represents the decay of sediment concentration with height above the surrounding substrate, $(\chi G)_0$ is the deposition rate for which the filtration rate is a maximum, and F_0 is the maximum filtration rate (Table 1).

The function f represents the proportion of the oysters that are not affected by sediment. For this study, we used a slightly different form for f than JC11. Because oysters grow at a range of elevations relative to the top of a reef, we define f as the fraction of the oyster layer not covered by sediment,

$$f = \begin{cases} 1, & S < B \\ \dfrac{O + B - S}{O}, & B < S < B + O \\ 0, & S > B + O \end{cases} \quad (6)$$

This difference in the form of f does not significantly alter the behavior of the model from that described by JC11, although the bifurcation point differs somewhat.

The solution of the equation set (Eqs. 1, 2, and 3) contains a bifurcation point, meaning it behaves in two distinctly different ways depending on the initial conditions. For this study, we used the same initial conditions as JC11: $O(0)$=0.01 m, $S(0)$=0.01 m, and the initial shell layer height $B(0)$ was varied. Therefore, the bifurcation point is expressed in terms of a threshold shell volume per unit plan area $B(0)$ and represents the minimum initial shell layer height above the surrounding substrate that is needed for the reef to survive. For small initial shell heights, below the threshold value, sediment accumulates more rapidly than the reef grows, the reef becomes covered by sediment, and the reef height eventually decays to zero (Fig. 2a). For larger initial shell heights, above the threshold value, reef growth exceeds the rate of sediment accumulation and the reef reaches a steady-state height at which the rate of growth is balanced by mortality (Fig. 2b). For the form of the model described above with the parameter values in Table 1, the threshold is $B(0)$=0.013 m. This value differs substantially from the value reported in JC11, primarily because we have opted to use a smaller shell degradation rate (γ=0.2 $year^{-1}$ rather than γ=0.7 $year^{-1}$), based on values observed by

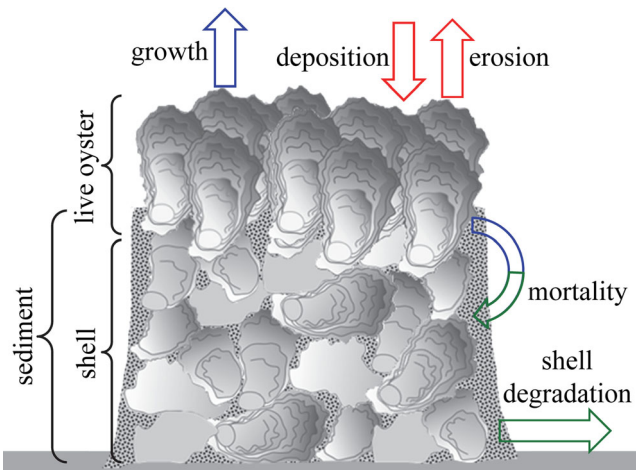


Fig. 1 Schematic diagram illustrating processes included in the model that affect live oyster volume (*blue*), shell volume (*green*), and sediment volume (*red*)

Table 1 Parameters in the Jordan-Cooley et al. (2011) model

Parameter	Description	Units	Value
r	Parameter controlling magnitude of logistic growth	year^{-1}	1
K	Oyster carrying capacity	m	0.3
μ	Mortality rate in the absence of sediment	year^{-1}	0.4
ε	Mortality rate of oysters covered in sediment	year^{-1}	0.94
γ	Oyster shell degradation rate	year^{-1}	0.2[a]
β	Sediment erosion rate	year^{-1}	0.01
χ	Maximum possible sediment deposition rate	m year^{-1}	0.02
η	Decay rate of sediment deposition with reef height	m^{-1}	3.33
F_0	Maximum sediment filtration rate	year^{-1}	1
$(\chi G)_0$	Deposition rate for which filtration is a maximum	m year^{-1}	0.02

Values given are those used in our simulations and match those used in the original study unless noted otherwise

[a] Value differs from that used by Jordan-Cooley et al. (2011). Original study used γ=0.7 year^{-1} . The value used in this study is based on Powell et al. (2006)

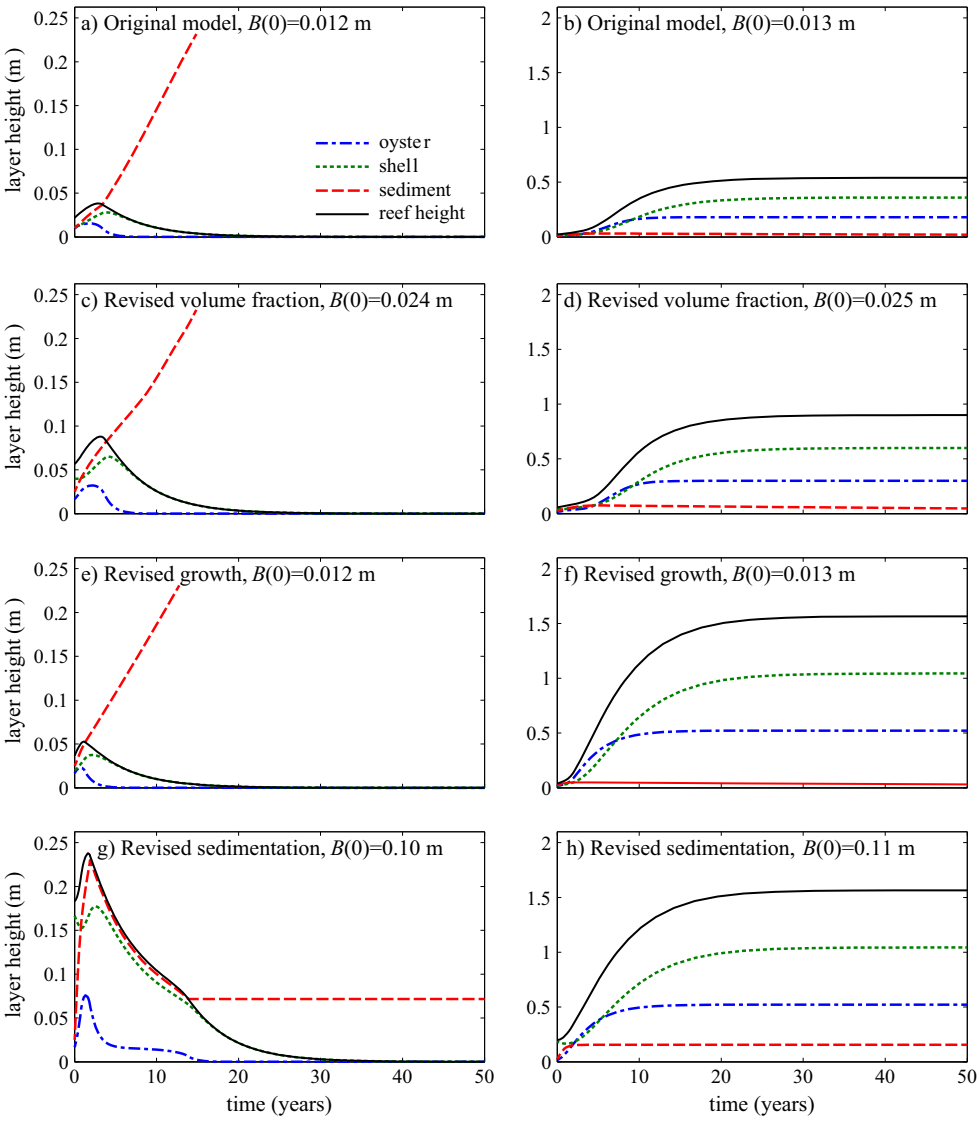


Fig. 2 Evolution of oyster, shell, and sediment layer heights after each model revision. In each case, the *left panel* shows model behavior for the largest initial reef height for which the reef dies, and the *right panel* shows the smallest initial reef height for which the reef survives. Results in each row include all prior model revisions

Powell et al. (2006). The remainder of this section describes the changes we made to the model and the effects of these changes on model behavior and results.

Correction for Volume Fraction

In JC11, oyster, shell, and sediment volumes per unit plan area were directly equated to layer heights. In reality, the presence of shell elevates the position of the sediment in the reef, since sediment can only occupy space not already taken up by shell. To correct the model for this, the volumes of oysters, shell, and sediment were converted to layer heights taking into account the volume fraction occupied by each component of the reef matrix. Dividing each volume (O, B, and S) by its respective volume fraction (ϕ_O, ϕ_B, and ϕ_S) yields the heights of each layer ($h_O=O/\phi_O$;$h_B=B/\phi_B$;$h_S=S/\phi_S$).

The volume fractions for each component were assumed to be constant throughout the reef. Since the fraction of oysters above the sediment is dependent upon the height of sediment within the reef, the factor (f), which represents the proportion of oysters affected by sediment was calculated using the new layer heights,

$$f = \frac{h_O + h_B - h_S}{h_O}, \qquad \frac{B}{\phi_B} < \frac{S}{\phi_S} < \frac{B}{\phi_B} + \frac{O}{\phi_O} \tag{7}$$

When a 3:2 ratio of oyster volume to sediment volume, and shell volume to sediment volume were used in the model ($\phi_O=\phi_B=0.6$, $\phi_S=0.4$), the threshold initial height changed from $B(0)=.013$ m to $B(0)=.025$ m (Fig. 2c, d). The threshold height increased because the same volume of sediment forms a deeper sediment layer if sediment only occupies a fraction of the total volume in the reef; i.e., the presence of shell and oysters elevates the position of the sediment within the reef. Therefore, the reef must have a higher initial height for reef growth to outpace sediment accumulation. These values for ϕ_O, ϕ_B, and ϕ_S were used for all subsequent model runs.

Improvement of the Growth Term

In JC11 (Eq. 1), growth is represented using a logistic growth expression with a carrying capacity. To understand whether a logistic model is appropriate for oysters, we developed a model that includes the processes that control growth of a filter feeder in a turbulent boundary layer. The new growth model is based on a time-dependent energy-flow model introduced by Powell et al. (1994). The governing equation in the Powell model is

$$P = A - R \tag{8}$$

Net production (P) is the rate of increase in biomass of a single oyster, including both somatic tissue (tissue forming the body of the organism) and reproductive tissue (gametes). Net production is expressed as the difference between the rate of energy uptake from feeding (assimilation, A) and the rate that energy is expended by the oyster (respiration, R). Therefore, in the revised model, the system is assumed to be self-seeding because the growth model includes both adult growth and juvenile recruitment, but does not distinguish these.

Assimilation is a function of the volume of water filtered by an individual oyster per unit time (filtration rate, Q), the ambient food concentration ($c_{f,ref}$), and a reduction in food availability (ψ) that represents the reduction in concentration at the top of the reef due to the net filtration by all the oysters on the reef,

$$A = 0.75 \; Q \; c_{f,ref}(1-\psi) \tag{9}$$

Multiplying the filtration rate by the ambient food concentration yields a rate of food uptake for an individual oyster. Filtration rate (Q) varies with oyster biomass and water temperature. Here, we used a constant filtration rate of 1 mL/s, a reasonable value for a medium sized (~8 cm) oyster at ~28 °C (Powell et al. 1994). The assimilation efficiency, or the fraction of food mass consumed that was assimilated by an oyster, was assumed to be 75 % (Powell et al. 1994).

Food availability to an individual oyster is reduced as oyster density (σ) increases, where σ is the number of oysters per unit plan area of reef. As oysters filter water, they remove food from the layer of water immediately above the reef, reducing the concentration in this layer. We assumed that the food supply in this layer is replenished primarily by vertical mixing. The balance between food depletion by filtration and food replenishment by vertical mixing results in a food concentration profile that increases logarithmically with height (z) above the reef (Monismith et al. 2010).

$$c_f(z) = c_{f,bed}\left[\frac{Q\sigma}{\kappa u_*}\ln\left(\frac{z}{z_{bed}}\right) + 1\right] \tag{10}$$

The ratio of food available at a given oyster density σ to the food that would be available in the absence of filter feeders is equivalent to the concentration at the top of the reef divided by the ambient food concentration $\frac{c_{f,bed}}{c_{f,ref}}$. Thus,

$$1-\psi = \frac{c_{f,bed}}{c_{f,ref}} = \left[\frac{Q\sigma}{u_*\kappa}\ln\left(\frac{z_{ref}}{z_{bed}}\right) + 1\right]^{-1} \tag{11}$$

We used $z_{ref}=1$ m and $z_{bed}=0.01$ m.

Respiration represents the rate at which energy is expended by the oyster. We assumed that the rate of energy consumption was proportional to the energy assimilated, with proportionality constant λ,

$$R = \lambda A \tag{12}$$

The net production of an individual oyster, obtained by combining Eqs. 8, 9, 10, 11, and 12, decreases with increasing total oyster volume due to the reduction in food availability (Fig. 3).

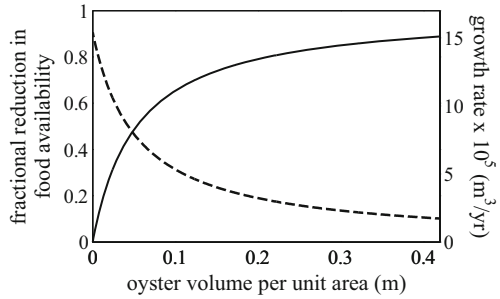


Fig. 3 *Solid line* Fractional reduction in food availability to an individual oyster due to filtering by all oysters on the reef as a function of total oyster volume. The curve asymptotes to unity (100 % reduction) for large oyster volumes. *Dashed line* Growth rate of an individual oyster versus total oyster volume on the reef

The Powell growth model (Eqs. 8, 9, 10, 11, and 12) describes the rate of change of biomass of a single oyster. The conversion from oyster biomass to oyster volume was accomplished in two steps: (1) converting from oyster biomass to oyster length and (2) converting from oyster length to oyster volume. To convert from biomass to length, we applied a curve fit to biomass and length data provided in Table 1 of Powell et al. (1994) (Fig. 4a). The conversion from length to volume was determined using our own measurements. Oyster shell halves (N=69) spanning the lengths of all the size classes reported in Powell et al. (1994) were collected. Each shell was filled with modeling clay and submerged in water in a large measuring cylinder. The shell volume was determined from the displacement of the water. Since only shell halves were used, measured volumes were doubled to calculate the total volume of each oyster. A curve fit was applied to the experimental data to obtain an expression that relates oyster length to oyster volume. Because there is a cubic relationship between length and volume, a cubic function was used for the fit (Fig. 4b).

The net production described above (Eqs. 8, 9, 10, 11, and 12) is the volume rate of change for a single oyster. The net production was therefore multiplied by the oyster density to determine the rate of change of oyster volume per unit area for the entire reef, yielding a new growth expression that replaces the logistic growth term in Eq. 1.

$$\frac{dO}{dt} = \underbrace{\frac{b_1 f\, O}{b_2 O + 1}}_{\text{growth}} - \underbrace{[\mu f + \varepsilon(1-f)]O}_{\text{mortality}} \tag{13}$$

In Eq. 13, parameters from Eqs. 8, 9, 10, 11, and 12 are combined into the new constants b_1 and b_2 to show the basic form of the new growth term. The constant b_1 represents the fractional rate of increase due to growth in the absence of competition, and b_2 is a constant that represents the density-dependent growth reduction due to food depletion by the collective filtering of all the oysters (Online Resource 1).

The two growth curves are compared in Fig. 5. In the JC11 growth term, once the oyster volume reaches the carrying capacity, the growth rate becomes zero. The revised growth model asymptotes to a constant growth rate at high oyster densities. This is a more realistic behavior for growth of a bed of filter feeders. When the mortality term is subtracted from the growth term, the new model has a similar shape to the terms from the JC11 model. In both cases, the net increase in oyster volume is zero for no oysters, achieves a maximum at an intermediate oyster volume, and crosses zero and becomes negative for large oyster volumes. However, the two models have different skewness; the net growth rate is higher at small oyster densities in the new model. The shape of the revised growth term is similar to the relationship between accretion rate and oyster density observed by Schulte et al. (2009).

When the JC11 growth term was replaced by the new growth term, the threshold initial shell layer volume per unit area changed from $B(0)$=0.025 m to $B(0)$=0.013 m (Fig. 2e, f). The threshold height decreased because the growth rate is larger for small oyster volumes in the revised model than in the original model. This value is smaller than the height of most individual oyster shells; thus, the reef effectively survives at all possible initial heights for these parameter values. Since each revision to the model was made successively, at this stage, the model includes the volume fraction correction but contains the original deposition and erosion terms.

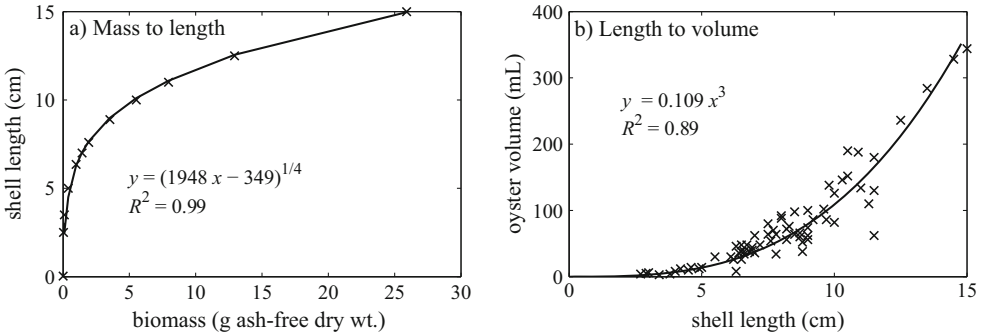


Fig. 4 Allometric relationships used to convert from oyster biomass to volume in the new growth model: **a** oyster biomass and shell length data and fit from Table 1 of Powell et al. (1994) and **b** our own measurements of oyster length and volume with cubic least-squares fit

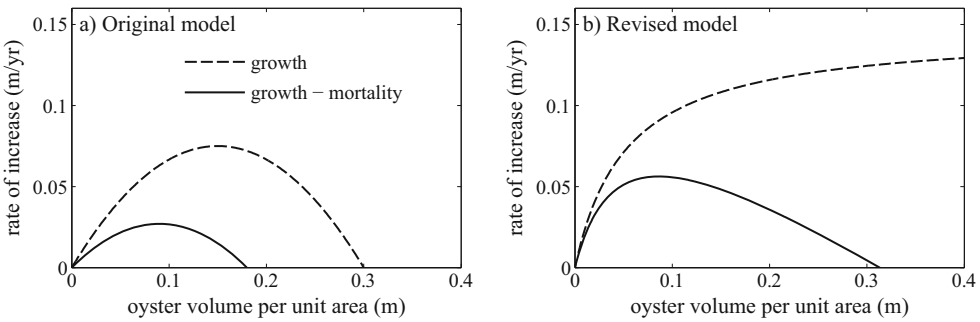


Fig. 5 Comparison of the growth rates in **a** the JC11 model and **b** the revised model. In each case the *dashed line* is the oyster growth term, and the *solid line* is the net growth which includes both growth and mortality

Improvement of the Sediment Deposition and Erosion Models

The net rate of sediment accumulation is equal to the rate of deposition minus the rate of erosion. In the JC11 sedimentation equation (Eq. 3), the rate of erosion is proportional to the total volume of sediment on the reef. Deposition was modeled by JC11 as a maximum possible deposition rate (χ) multiplied by a factor (G) that decays exponentially as reef height increases and a second factor that decreases exponentially as oyster filtration increases. While parameter values in the erosion and deposition terms (erosion rate β, maximum sediment deposition rate χ) were based on field measurements in the York River estuary, there is no way to vary flow and sediment properties in the original model.

Sedimentation is a physical process that depends on factors such as shear stress exerted by the flow on the top of the sediment layer, the sediment settling velocity, and the sediment concentration profile in the water column. In the revised model, deposition and erosion were modified to better represent these processes. Although our representations of deposition and erosion are necessarily simple and do not account for processes such as flocculation or details of how flow is modified by the reef structure, the revised terms are a first-order representation of the processes of deposition and erosion. We assume that sediment is noncohesive. We also assume that the flow over the oyster reef is well represented as a steady current; expressions for E and D would need to be modified to model processes under waves.

For the erosion term (E), we use the function (e.g., Warner et al. 2008),

$$E = E_0\left(\frac{\theta-\theta_c}{\theta_c}\right), \quad \theta > \theta_c \tag{14}$$

Here, E_0 is a bed erodibility constant, $\theta=\tau_{bed}/[(\rho_s-\rho_w)gd]$ is the Shields parameter, θ_c is the critical Shields parameter, ρ_s is the sediment density, ρ_w is the density of water taken to be 1000 kg/m^3, g=9.8 m/s^2 is acceleration due to gravity, d is sediment grain size, and ν is the kinematic viscosity of water taken to be 10^{-6} m^2/s. Erosion occurs when the Shields parameter, θ, exceeds the critical Shields parameter, θ_c, the smallest value for which shear stress is sufficient for sediment resuspension. When $\theta<\theta_c$, no erosion occurs (E=0). The critical Shields parameter varies with grain size and cohesion, and appropriate values will vary depending on sediment properties at a site (e.g., Miller et al. 1977). A value of θ_c=0.05 was used for all model runs in this paper. We used a single value for the current and effective grain size for each simulation in order to investigate how oyster reef development varies with these parameters.

When the top of the sediment layer is above the top of reef, the shear stress is set to a constant value determined by the flow speed,

$$\frac{\tau_{bed}}{\rho_w} = u_*^2 = C_{D,reef}U^2 \quad , h_S > h_O + h_B \tag{15}$$

where u_* is the friction velocity, U is the current above the reef (nominally at 1 m), and $C_{D,reef}$ is the drag coefficient on the reef. When the top of the sediment layer is below the top of the reef, the shear stress exerted on the sediment is smaller because the reef structure exerts a drag force on the water that reduces current below the top of the reef. Whitman and Reidenbach (2012) measured currents and turbulence above and between vertically oriented oysters on a reef and observed that shear stress decays exponentially within the reef structure. Similar shear stress profiles have been observed within other types of submerged obstacle arrays such as vegetation canopies (e.g., Ghisalberti and Nepf 2004; Finnigan 2000). Consequently, the closer the top of the sediment layer is to the top of the reef, the more erosion occurs. To represent this in our model, when the top of the sediment layer is below the top of the reef, the shear stress acting on the sediment (τ_{bed}) is set to the shear stress at the top of the reef multiplied by a factor that decays exponentially with increasing distance below the reef top,

$$\frac{\tau_{bed}}{\rho_w} = u_*^2 e^{-\frac{h_O+h_B-h_S}{m}} = C_{D,reef}U^2 e^{-\frac{h_O+h_B-h_S}{m}}, h_S < h_O + h_B \tag{16}$$

The decay scale m was set to 0.02 m to be consistent with the results of Whitman and Reidenbach (2012). Whitman and Reidenbach (2012) observed $u_*^2 \sim 0.01U^2$ at the top of the oyster reef in their study (their Fig. 5),

corresponding to $C_{D,reef}$~0.01. This value was used for all of our simulations (Table 2).

The erosion term in the revised model is controlled by the height of the sediment layer relative to the reef top (Fig. 6). In the JC11 model, the magnitude of erosion increases linearly with increasing sediment volume and does not vary in response to the position of the sediment within the reef structure. The revised erosion term decays exponentially as the sediment level becomes further below the top of the reef, simulating the sheltering effect of the reef structure. The erosion rate in the revised model is also larger when flow speed is faster.

In the revised model, we also used a process-based representation of deposition. The rate of deposition is equivalent to the flux of sediment to the bed due to settling and is equal to the sediment concentration at the reef top $\left(c_{s,h_O+h_B}\right)$ multiplied by the settling velocity (W_s).

$$D = c_{s,h_O+h_B} W_s \tag{17}$$

The settling velocity was assumed to be related to grain size (d) by Stokes law and is a function of the acceleration due to gravity (g), the sediment density (ρ_s), the density of water (ρ_w), and the kinematic viscosity (ν) of water,

$$W_s = \frac{\rho_s - \rho_w}{\rho_w} \frac{g d^2}{18\nu} \tag{18}$$

Stokes law gives good estimates of settling velocity for well-rounded particles with d<0.1 mm, but overestimates settling velocities by a factor of 2 for particles with d=0.2 mm (Dietrich 1982). Stokes law is used here for the complete range of grain sizes (0.01–0.2 mm) for illustrative purposes. When using the model to site and design reefs, it may be preferable to use direct measurements of sediment settling velocity rather than grain sizes. In the revised model, we assumed that the balance between downward sediment settling and upward turbulent mixing results in a Rouse sediment concentration profile adjacent to the reef (van Rijn 1993).

The sediment concentration profile was assumed not to change significantly when water meets the reef. The sediment concentration at the top of the reef is therefore the concentration at height $z=h_O+h_B$ above bottom from the Rouse profile over a flat bottom,

$$c_{s,h_O+h_B} = c_{s,a}\left(\frac{H-(h_O+h_B)}{h_O+h_B}\cdot\frac{a}{H-a}\right)^{\frac{W_s}{\kappa u_*}} \tag{19}$$

where $c_s$is the concentration at $z=h_O+h_B$, $c_{s,a}$ is the concentration at some reference height a very close to the bed, H is the total water depth, and κ=0.4 is the Von Karman constant. The friction velocity was evaluated using $u_*^2 = C_{D,sand}U^2$, with $C_{D,sand}$=0.002 to represent the value for the flat sandy bottom adjacent to the reef. The reference concentration $c_{s,a}$ was computed from a fixed depth-

Table 2 Parameters in the revised model and default values used for the simulations

Parameter	Description	Units	Value
ϕ_O	Volume fraction occupied by oysters	–	0.6
ϕ_B	Volume fraction occupied by shell	–	0.6
ϕ_S	Volume fraction occupied by sediment	–	0.4
μ	Mortality rate in the absence of sediment	year^{-1}	0.4
ε	Mortality rate of oysters covered in sediment	year^{-1}	0.94
γ	Oyster shell degradation rate	year^{-1}	0.2
Q	Individual oyster filtration rate	m^3 s^{-1}	1×10^{-6}
$c_{f,ref}$	Ambient food concentration	kg m^{-3}	1×10^{-4}
M	Mean biomass (dry weight) of an individual oyster	kg	3×10^{-4}
V	Mean volume of an individual oyster	m^3	5.6×10^{-5}
λ	Ratio of respiration to assimilation	–	0.67
d	Sediment grain size	m	1×10^{-4}
ρ_s	Density of sediment grains	kg m^{-3}	2650
$C_{D,\ reef}$	Drag coefficient relating shear stress at reef top to current	–	0.01
$C_{D,\ sand}$	Drag coefficient for flat bottom adjacent to reef	–	0.002
E_0	Bed erodibility constant	m s^{-1}	1×10^{-6}
θ_c	Critical Shields parameter	–	0.05
U	Current speed at 1 m above reef top	m s^{-1}	0.1
m	Decay rate of shear stress with depth below top of reef	m^{-1}	0.02
$c_{s,davg}$	Depth-average volumetric sediment concentration	–	4×10^{-6}
H	Depth from reef base to water surface	m	5

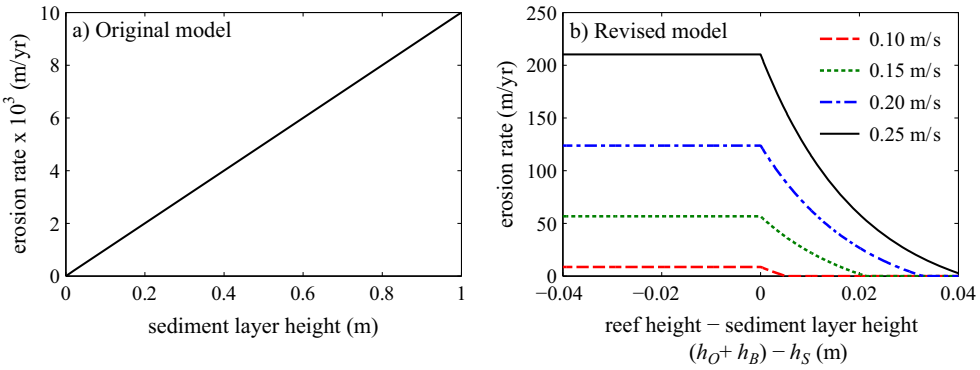


Fig. 6 Comparison of **a** original and **b** revised erosion terms. In **a**, the *x*-axis is the height of the sediment layer within the reef relative to the substrate surrounding the reef. In **b**, the *x*-axis represents the position of the top of the sediment layer relative to the top of the reef. Results are shown for four different flow speeds

averaged sediment concentration $c_{s,davg}$ by integrating Eq. 19 over the water depth,

$$c_{s,a} = c_{s,davg}\left[\frac{1}{H}\int_a^H \left(\frac{H-(h_O+h_B)}{h_O+h_B}\cdot\frac{a}{H-a}\right)^{\frac{W_s}{\kappa u_*}} dz\right]^{-1} \quad (20)$$

An intermediate value of total suspended sediment concentrations measured by Fugate and Friedrichs (2002) in the Chesapeake Bay was used for $c_{s,davg}$ (Table 2). The primary factors controlling sediment deposition onto the reef are therefore grain size, sediment density, and water velocity (Eqs. 17, 18, 19, and 20, Fig. 7). Because the sediment concentration profile was based on depth-averaged sediment concentration, the revised deposition term can account for both sediment that is resuspended locally and sediment from runoff.

The revised deposition model does not include a reduction to deposition from oyster filtration because sediment taken up by oysters is deposited on the reef as pseudofeces and feces. Because the sediment particle diameter is assumed to be constant for a given simulation, the settling velocity is a constant. Consequently, changes in sediment deposition with time are controlled by the location of the top of the reef within the Rouse concentration profile. The shapes of the deposition versus reef height curves in the JC11 and revised models are similar (Fig. 8); however, the revised sedimentation terms enable the model to be connected to conditions on a real reef.

At this stage, the JC11 sedimentation terms were replaced with the new forms of the deposition and erosion equations, completing the model revisions (Online Resource 1). During the initial phases of the model development, the numerical simulations were run using MATLAB's ode45 solver. However, implementing the revised erosion and deposition terms resulted in a stiff equation set and solutions that varied rapidly with time. The equations are stiff because the erosion (E) and deposition (D) terms are much larger in magnitude than the difference between the two terms (dS/dt). This characteristic is inherent to sediment dynamics and is one of the reasons that sediment dynamics are so difficult to accurately model. Consequently, the stiff equation solver ode23s was used to perform the model runs included in this paper. While ode23s has a lower order of accuracy than ode45, it improves model runtime and eliminates unstable and unrealistic solution behavior.

Changing the sediment deposition and erosion terms in the model moved the threshold initial reef height from $B(0)$= 0.013 m to $B(0)$=0.11 m (Fig. 2g, h). This increase in the threshold height is likely because sediment deposition is more rapid; therefore, the oyster bed must start higher above the surrounding sediment for growth to outpace sediment accumulation. Additionally, in the revised model, erosion does not remove sediment from the reef until the shear stress exceeds the critical shear stress, which only occurs once sediment is within centimeters of the top of the reef. This is quite different from JC11 where the erosion rate was proportional to the total volume of sediment, even if the sediment layer was well beneath the top of the reef. This change in the treatment of erosion represents a fundamental change to the model that alters its behavior, as described further below.

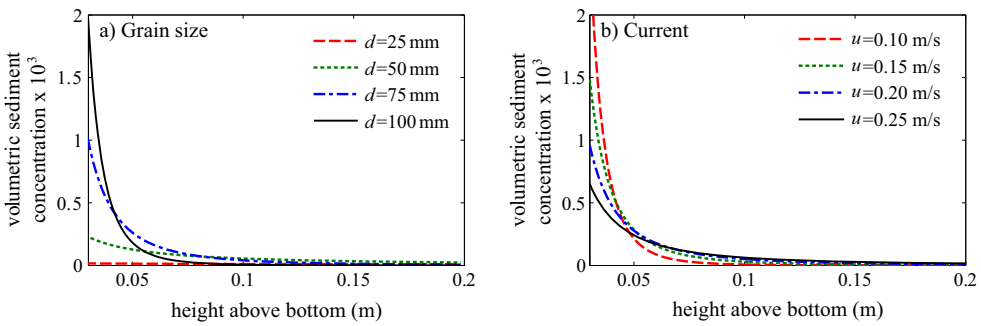


Fig. 7 Example Rouse sediment concentration profiles used for the revised deposition model. *Curves* are shown for four different **a** grain sizes and **b** flow speeds

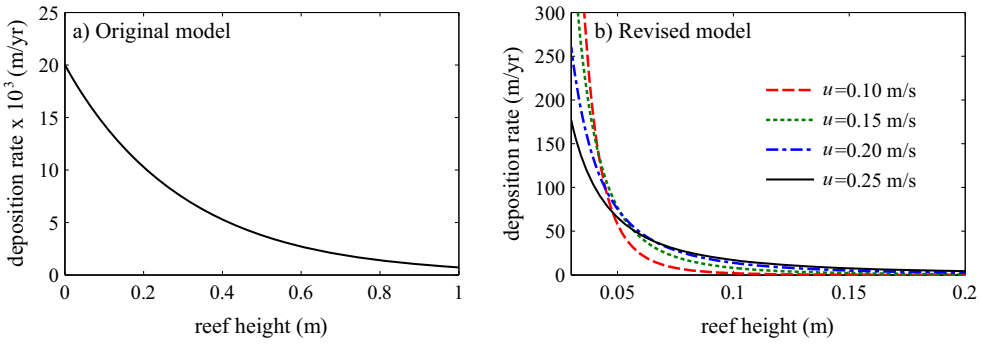


Fig. 8 Comparison of **a** the JC11 deposition term and **b** the revised deposition term versus height of reef above surrounding substrate. Curves are shown for four different flow speeds

Sensitivity of Model Results to Environmental Parameters

After the model was revised, we investigated the sensitivity of the results to model parameters. Parameters in the growth, erosion, and deposition terms were varied. First, parameters were varied individually while keeping all other parameters constant. We examined the effect of these parameters on the final stable-state reef height, defined as the reef height at the end of the model run (t=50 years). This process was repeated for five different initial reef heights $B(0)$ across a range typical for oyster reef restoration projects (Schulte et al. 2009; Stokes et al. 2012). To investigate the interacting effects of different parameters, simulations were then repeated while varying two parameters and keeping all other parameters constant. Water velocity and sediment diameter were varied together because these parameters control the rate of sediment accumulation. Ambient food concentration and shell degradation rate were varied together because these parameters control the net rate of change of reef height. Stable-state reef height was selected as the output variable since high stable-state reef heights indicate that the parameter value, or combination of parameter values, is conducive to successful reef restoration.

Dependence of Final Reef Height on Flow Speed

Water velocity was varied over the range of 0–0.25 m/s, representative of currents in shallow estuarine environments where oyster reefs live. The dependence of final reef height on water velocity is complex because water velocity affects the growth term, the deposition term, and the erosion term. For small flow speeds, increasing the water velocity increases the stable-state reef height because vertical mixing (u_*) is increased and food supply at the top of the oyster bed is replenished more rapidly (Eq. 11, Fig. 9a). However, for each initial reef height $B(0)$, the final reef height is zero for a range of intermediate velocities (0.025–0.075 m/s) because as vertical mixing (u_*) increases, sediment is distributed higher in the water column and deposition onto the oyster reef is increased. When the rate of deposition exceeds the initial rate of reef growth, sediment accumulates and the oyster bed becomes buried by sediment. This corresponds to the threshold initial reef height, described previously. When the initial reef height is larger, this transition in model behavior occurs at a faster velocity because more mixing is required to distribute the sediment high enough in the water column to smother the reef (Figs. 7b and 9a). The final reef height is nonzero for velocities >0.1 m/s because the shear stress exceeds the critical shear stress, erosion keeps the sediment level below the top of the reef, and a live oyster layer is maintained. In this regime, the final reef height increases as the current increases for two reasons. Firstly, larger currents result in greater erosion and hence a thicker layer of live growing oysters at the reef top. Secondly, larger currents result in greater vertical mixing and hence a larger food supply and faster growth rate.

Dependence of Final Reef Height on Sediment Grain Size

Sediment diameter was varied over the range of 0.01–0.2 mm, which represents grain sizes from silt to medium-sized sand. For very small grain sizes, the reef attains its maximum steady-state height because deposition is negligible. As grain size increases, the final reef height decreases firstly because deposition becomes significant and secondly because the critical shear stress for erosion increases as grain size increases and the vertical level within the reef at which $\theta>\theta_c$ gets closer to the reef top. For grain sizes d>0.08 mm, $\theta<\theta_c$ even at the reef top; therefore, erosion does not occur and the reef is buried by sediment. When the grain size is larger still, sediment is more concentrated near the bottom (Fig. 7a); therefore, the sediment concentration at the reef top decreases as grain size increases. Beyond a critical grain size, deposition is small enough that reef growth outpaces sediment accumulation and the reef is able to survive. This corresponds to the threshold initial reef height described previously. The grain size at which this transition occurs is smaller for larger initial reef heights and occurs at grain sizes in the large silt to small sand range (0.08–0.13 mm). This is why reefs with larger initial heights are able to survive in environments with smaller grain sizes.

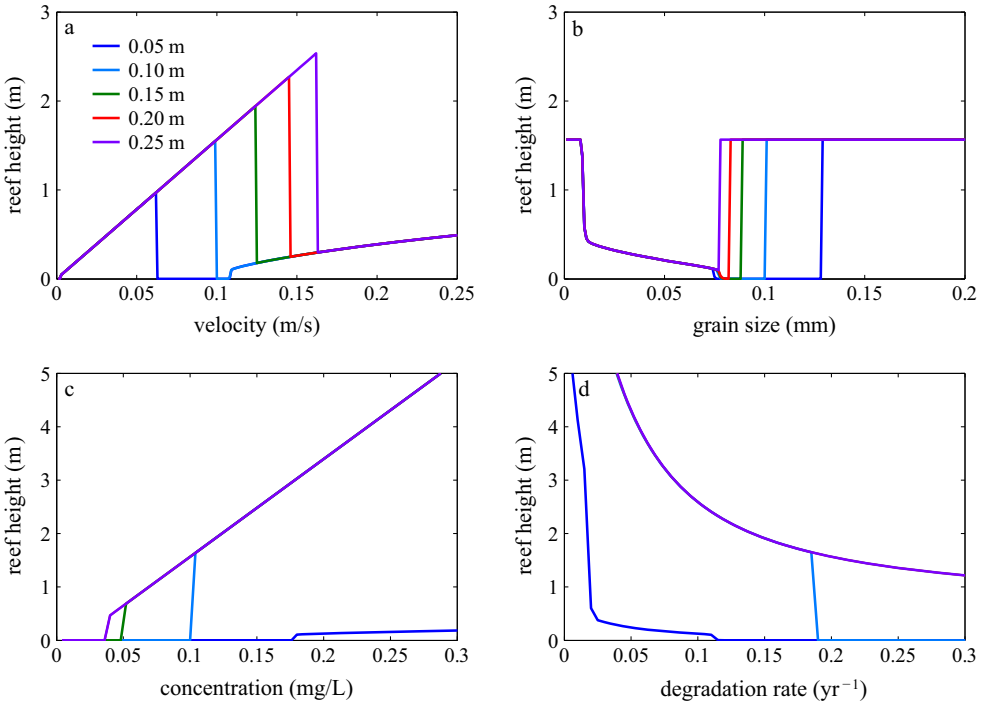


Fig. 9 Response of final reef height (t=50 years) to different parameters values: **a** flow speed, **b** grain size, **c** ambient food concentration, and **d** shell degradation rate. In each case, all other parameters were held constant (Table 2 values). *Colors* represent different initial reef heights. Larger final reef heights indicate that parameter values are more suitable for reef restoration

Dependence of Final Reef Height on Food Concentration

Ambient food concentration was varied from 0 to 0.4 mg/L (Fig. 9c), similar to the range in Powell et al. (1994). These values correspond to organic matter in seston measured in Galveston Bay and Chesapeake Bay when there are no plankton blooms (Soniat et al. 1984; Berg and Newell 1986). At low ambient food concentrations, the reef is not able to survive because the rate of sediment accumulation exceeds the rate of net growth. As food concentration is increased, the initial growth rate increases until the rate of reef growth exceeds the rate of sediment accumulation. At this food concentration, there is a jump in final reef height corresponding to the threshold initial reef height described previously. As ambient food concentration is increased further, the final reef height increases because the energy available to the oysters for growth increases linearly with ambient food concentration (Eq. 9). For the smallest initial reef height (0.05 m), a much larger ambient food concentration is required for the net growth rate to outpace sedimentation and for the reef to survive than for the larger reef heights because the sediment concentration at 5 cm above bottom is very high. For this reef, growth did not outpace sedimentation for the range of food concentrations tested; however, a reduced steady-state reef height was achieved because a live oyster layer was maintained by erosion.

Dependence of Final Reef Height on Shell Degradation Rate

Shell degradation rate was varied from 0 to 0.4 year^{-1}, based on the range observed by Powell et al. (2006). The dependence of final reef height on shell degradation rate is the inverse of the dependence on ambient food concentration (Fig. 9d). When the shell degradation rate is small, the final reef height is large. As the shell degradation rate increases, the stable-state reef height at which growth is balanced by degradation decreases. Above a critical degradation rate (0.2–0.3 year^{-1}), the reef top is so low in the water column that the rate of deposition exceeds the net rate of growth and the reef is buried by sediment. This corresponds to the transition from model behavior where the initial reef height is above the threshold value to model behavior where the initial reef height is below the threshold. The lower the initial reef height is, the smaller the critical degradation rate is . Intuitively, slower degradation rates result in taller more robust reefs.

Dependence of Final Reef Height on Flow Speed and Sediment Grain Size

Here, we investigate interactions between grain size and water velocity that affect reef restoration success. Varying grain size conjunctively with the water velocity resulted in three distinct outcomes: (1) The sediment concentration is high at the top of the reef so deposition outpaces oyster growth and the reef is buried by sediment; (2) deposition is high relative to oyster growth but the flow speed is fast enough that the shear stress exceeds the critical shear stress and erosion maintains a live oyster layer; or (3) the sediment concentration at the reef top is sufficiently low that deposition is negligible and the reef survives (Fig. 10a).

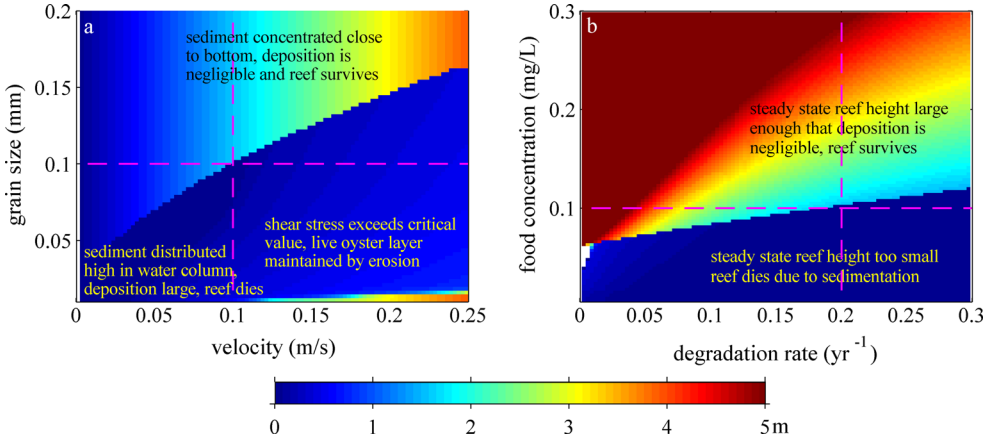


Fig. 10 Final reef height (t=50 years) versus two environmental parameters that were varied jointly. Values shown are for an initial reef height $B(0)$=0.1 m. Higher final reef heights indicate environments more suitable for reef restoration. The sharp transition in model behavior that runs across the middle of each panel corresponds to the threshold initial reef height. *Magenta lines* correspond to the cyan curves in Fig. 9 where a single parameter was varied

When the sediment grain size is <0.1 mm (clay and fine silt) and the water velocity is <0.1 m/s, the reef does not survive. This is because small particles settle slowly, and, even with relatively little mixing (small flow speeds), the sediment is distributed high in the water column. The sediment concentration at the top of the reef is high, and there is heavy deposition onto the reef but no erosion. For each grain size, there is a corresponding flow speed above which the development of the reef is negatively impacted (buried or reef height drastically reduced) by sedimentation. This critical flow speed increases as grain size increases because more vertical mixing is required to mix larger particles high in the water column.

Even when reef development is negatively impacted by sedimentation, if the shear stress exceeds the critical shear stress, a small population of live oysters is maintained. When the shear stress at the reef top exceeds the critical shear stress, erosion removes a portion of the sediment from the top of the reef, enabling a layer of live oysters to survive. However, since the reef structure protects sediment within the reef from erosion, only sediment from the top few centimeters of the reef is removed (Fig. 6b). The net growth rate is therefore reduced, and the final reef height is much smaller than a reef that is unaffected by sedimentation.

If the flow speed is less than the critical value for a given grain size, the sediment concentration at the reef top is small, reef growth outpaces sediment deposition, and the reef survives. In this regime, the stable-state reef height increases with flow speed due to increased food availability and hence larger growth rates. The maximum stable-state reef height occurred when the water velocity was its maximum value (0.25 m/s), and the grain size was larger than the critical value for that flow speed (0.18 mm). Overall, large grain sizes coupled with fast flow velocities provide better environments for reef restoration.

Dependence of Final Reef Height on Food Concentration and Shell Degradation Rate

Jointly varying the ambient food concentration and the shell degradation rate resulted in two different outcomes (Fig. 10b): (1) The net growth rate (growth minus degradation) is smaller than the rate of deposition and the reef is buried, or (2) the net growth rate is fast enough and the steady-state reef height is large enough that growth outpaces deposition and the reef survives. For each shell degradation rate, there is a food concentration below which the reef is unable to survive. The ambient food concentration determines the rate of growth; thus, at low ambient food concentrations, the reef does not grow rapidly enough to escape sedimentation. The critical food concentration increases with shell degradation rate because reef survival depends on the reef top being high in the water column, above the concentrated sediment layer. The more rapidly the reef structure degrades, the faster the oysters must grow in order to escape burial. The tallest reefs occurred when shell degradation was low and the ambient food concentration was high since this represents the combination of parameters for which the reef loses mass most slowly and grows most rapidly.

Discussion

In real oyster reefs, net reef growth is affected by many processes that occur simultaneously and differ between sites, making mechanisms leading to reef restoration success or failure difficult to isolate. The model developed in this study includes mathematical representations of physical processes and contains measurable environmental parameters. It can therefore be compared with, and aid interpretation of, field measurements. The model produces three distinct possible

outcomes depending on the combination of environmental parameters and initial reef height: (1) reefs that die due to smothering by sediment, (2) low-density reefs that are negatively impacted by sediment, and (3) reefs where deposition is negligible and the reef thrives. These states are consistent with findings of the field study conducted by Schulte et al. (2009). In the Schulte study, low relief reefs either died or survived with low oyster density, and high relief reefs flourished. Our model is also consistent with the Schulte et al. study in that high relief reefs, or reefs with a larger initial reef height, are more resilient. Taller reefs survive in environments with higher water velocities, smaller grain sizes, lower food concentrations, and faster shell degradation rates than reefs with lower initial heights (Fig. 9). The threshold initial reef heights and final reef heights predicted by the model are realistic for typical environmental parameter values. Although the Schulte study was only conducted over a five year period and water velocities and grain sizes were not reported, some of their low relief (8–12 cm tall) reefs survived and others did not, suggesting a threshold value close to this range.

The new physically realistic growth and sedimentation terms introduce a large number of new parameters that must be set in the revised model. The sensitivity analysis enabled us to assess the response of the model to variations in these parameters. The sensitivity analysis also allowed us to determine how different environmental parameters affect reef restoration success in order to predict the optimal combination of parameters for reef restoration.

Our results suggest that reefs are unable to survive when ambient food concentrations are too low because sediment accumulates more rapidly than the reef grows and that steady-state reef height increases as food concentration increases. The growth of an individual oyster is linearly proportional the food concentration, the organism's filtration rate, and the efficiency with which food energy is assimilated (Eq. 12); therefore, reef survival and final reef height vary in a similar way with all of these parameters. A reef containing organisms with slower filtration rates or lower efficiencies would be less likely to survive and would have lower final reef heights than more efficient organisms and organisms with faster filtration rates. While the general behavior of the growth model is realistic, results from the sensitivity analysis suggest that it might be too sensitive to the ambient food concentration. For initial reef heights above the threshold for survival, the final reef height increases linearly with food concentration; a 0.2-mg/L increase in food concentration results in a 2-m increase in final reef height. Such a small increase in food availability is unlikely to have such a large impact on reef height. This issue is not resolved by capping the assimilation rate for an individual because, at the steady-state reef height, the density-dependent growth term controls the reef growth rate and individual oyster assimilation is limited by food supply. In this part of the parameter space, it is likely that other processes not included in the model (e.g., proximity to the water surface, predation) control the steady-state reef height.

Steady-state reef heights are smaller when the shell degradation rate is larger because shell loss due to degradation balances shell gain due to mortality at a lower value of B (Fig. 9d). Furthermore, for higher shell degradation rates, a larger initial shell layer height is needed in order for the reef to survive. Reefs are expected to be less successful in environments where shell degradation is rapid, suggesting that sedimentation may pose an even greater problem in the lower pH estuaries of the future. Powell et al. (2006) found shell degradation rates were most rapid at sites in the intermediate salinity region of the Delaware Bay, and Waldbusser and Steenson (2011) found shell degradation to be most rapid at low pHs. This suggests that restoration efforts will be more successful in environments with a pH of >7 (Waldbusser and Steenson 2011) and restored in the upper or lower portion of an estuary to avoid intermediate salinities.

The increase in steady-state reef height with increasing sediment grain size, or equivalently, faster settling velocity, suggests that restored reefs are likely to have fewer sedimentation problems in environments where restored substrate is surrounded by sand as opposed to clay or silt (Fig. 9b). Model results suggest that optimal reef restoration sites have large grain sizes coupled with fast water velocities. Under these conditions, the large grain size keeps the sediment concentration at the reef top very small, and the fast velocity rapidly replenishes the food supply to oysters on the reef. Although we have not found previous studies that report both grain size and flow speed at the same restoration site, Rodriguez et al. (2014) observed the major fraction of sediment to be in the sand size class (>0.06 mm) at the reef base in cores of restored intertidal reefs in North Carolina. At another successful intertidal restoration site in Virginia, Whitman and Reidenbach (2012) observed flow speeds of 20–30 cm/s during rising and falling tides. Although many of the processes on intertidal reefs are not represented in our model, it does produce results that are consistent with the ranges observed in these studies.

The sensitivity analyses provided insight into site characteristics conducive to successful reef restoration. This information could be coupled with measurements at potential oyster reef restoration sites to aid site selection and inform restored reef design. Measurements of current speeds (v), total suspended solids ($c_{s,davg}$), chlorophyll-a concentrations ($c_{f,ref}$), and grain size distributions (d) or settling velocities (W_s) at potential sites could be used as input parameters for the model. The relative success of oyster reefs at potential sites could then be assessed based on reef heights and live oyster volumes at the end of the simulations. Once a site has been selected, sediment concentration profiles, their variation in time, and extreme values could be estimated from measured flow conditions and sediment properties (Eq. 19). Reefs could be designed with initial heights above the part of the water column where sediment is highly concentrated. Although sedimentation parameters are the most

important for determining reef survival, growth parameters are the most important for determining reef productivity. Therefore, once a group of sites with appropriate sedimentation characteristics has been determined, the sites with the greatest food availability should be selected to maximize reef productivity and ecosystem services. By adjusting parameters in the growth term, the model could also be used to inform restoration projects for other species of bivalves.

The model described in this study represents the reef as a single, horizontally uniform unit. Real reefs are three-dimensional, and both the flow characteristics and sediment profiles are altered as water moves across the reef structure. Future work could extend the model to one and two dimensions and examine the effect of horizontally varying processes on reef development. Additionally, in this study, we assumed that flow conditions and sediment concentration profiles were constant in time. Flow conditions vary substantially in natural systems and consist of regularly varying tidal flows, irregular wind-driven currents, and surface waves. In turn, the flow variations affect sediment concentration profiles and erosion rates; therefore, future work should also investigate the effects of waves and temporally varying flows on net sedimentation rates and reef development. The model could also be modified to investigate the effects of processes not considered in this study such as periodic removal of oysters due to commercial harvesting.

Although models can always be further refined to improve their representation of natural processes, the simple model described in this paper provides a first order description of the effects of sediment dynamics on oyster reef development. The model has helped build understanding of how sedimentation affects reef restoration success. The sensitivity analysis has shown that the minimum initial reef height needed for a restored reef to succeed varies with measurable parameters such as sediment grain size or settling velocity, flow speed, and food concentration. The model is a useful tool for predicting the relative success of restored oyster reefs under different environmental conditions and illustrates the importance of assessing physical conditions at potential sites prior to commencing restoration projects.

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